

Lab 2

DATA 422: Advanced Data Science Methods and Ethics

Ungraded. **Paper quiz** Friday from the [Quiz 2 study bank](#) (Ch. 2 + Lab 2).

Open in Colab

[Open in Colab](#)

Notebook: [lab-02.ipynb](#). CPU runtime.

What you will do

1. **JIT OOP** + `nn.Module` — `class`, `self`, `__init__`; then fit $\sin x + \text{noise}$; plot data, true curve, and prediction; change depth, width, or activation from a config block.
2. **Heads** — same skeleton for numeric (MSE), **binary** (one logit + `BCEWithLogitsLoss`), and **three-class** (softmax + `cross_entropy`).
3. **Ch. 2 text** — word tokenization ([theLMbook](#) style), `nn.Embedding`, next-token loss, **perplexity**.

Wildfire Model 1 is on the [Project 1 stepping stone](#) (due Wed Sep 9), not in this lab.

Before you leave

Work the [Quiz 2 study bank](#) before Friday's paper quiz.